

ROSTOCK LAAGE, Germany

WMO: 101720

Lat: 53.9156N

Lon: 12.2789E

Elev: 194

StdP: 14.59

Time Zone: 1.00 (EUC)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.2	17.4	8.0	8.3	14.4	13.3	10.9	18.8	29.2	42.2	26.0	41.9	6.8	90	0.588

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.6	84.3	66.6	80.7	65.7	77.1	64.0	68.9	79.9	67.0	77.4	65.4	74.5	9.3	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.8	92.5	72.3	63.3	87.6	70.7	62.0	83.7	69.3	33.0	79.3	31.6	77.3	30.3	74.6	75.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.2	20.2	17.8	7.5	91.1	6.9	3.4	2.6	93.5	-1.5	95.5	-5.3	97.4	-10.3	99.8		
(5)			7.1	72.7	6.8	1.6	2.2	73.8	-1.8	74.8	-5.6	75.7	-10.5	76.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.5	32.9	34.7	38.9	47.2	54.5	60.3	64.5	64.4	57.7	49.4	41.0	35.3
	DBStd	13.06	8.24	7.79	6.92	6.87	6.32	5.53	5.38	5.19	5.15	6.21	6.29	8.03
	HDD50	2284	529	430	347	131	25	1	0	0	3	87	275	456
	HDD65	6181	994	850	809	533	332	163	76	75	226	483	720	920
	CDD50	1732	0	0	3	49	164	308	451	448	234	69	6	0
	CDD65	154	0	0	0	0	6	21	62	58	7	0	0	0
	CDH74	1506	0	0	0	15	79	250	587	506	68	1	0	0
	CDH80	379	0	0	0	1	13	65	159	133	8	0	0	0
Wind	WSAvg	8.4	9.2	9.4	9.6	8.9	8.3	8.2	8.0	7.4	7.7	7.9	8.0	8.7
Precipitation	PrecAvg	24.4	1.9	1.6	1.6	1.3	2.1	2.6	2.6	2.5	2.2	1.9	2.0	2.2
	PrecMax	31.4	3.7	3.5	4.0	2.8	4.1	5.4	11.3	5.7	6.3	3.4	4.6	4.7
	PrecMin	17.1	0.0	0.3	0.3	0.1	0.5	0.1	0.2	1.1	0.5	0.8	0.2	0.6
	PrecStd	3.5	1.0	0.7	0.8	0.7	0.9	1.1	2.0	1.1	1.3	0.7	0.9	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.2	53.9	62.0	75.7	81.4	86.4	89.3	88.1	80.7	69.4	58.2	53.6
		MCWB	48.8	48.3	51.4	59.6	65.0	68.4	69.0	67.9	64.7	60.8	54.7	50.3
	2%	DB	48.5	50.2	56.8	68.0	75.3	80.7	84.3	83.9	75.4	64.4	55.0	50.4
		MCWB	46.1	46.5	48.6	55.8	62.1	65.7	67.2	65.8	62.6	58.1	51.8	48.3
	5%	DB	46.2	47.9	52.2	64.1	71.3	75.6	80.6	79.7	71.2	61.0	52.1	48.1
		MCWB	44.1	44.8	46.6	53.0	59.2	63.2	66.2	65.3	60.9	56.0	49.4	46.1
	10%	DB	44.0	45.1	48.6	59.8	67.4	71.7	76.8	75.5	66.9	58.5	49.8	45.3
		MCWB	42.2	42.2	44.5	51.1	57.3	61.4	64.3	63.8	59.0	54.6	47.6	43.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.6	49.5	52.8	60.2	66.6	70.2	71.8	70.9	66.4	61.9	55.8	51.8
		MCDB	51.8	51.6	59.0	72.1	78.1	81.5	84.8	83.5	76.4	68.1	57.5	53.8
	2%	WB	46.3	47.1	49.7	56.8	63.6	67.2	69.1	68.2	63.9	58.8	52.3	48.6
		MCDB	48.0	49.6	54.4	66.5	72.7	78.1	80.2	78.4	72.3	63.2	54.6	50.0
	5%	WB	44.0	44.7	47.1	54.2	60.6	64.9	67.2	66.3	62.0	56.7	49.8	45.9
		MCDB	45.6	47.4	51.7	62.0	69.1	73.4	77.9	75.7	69.1	60.4	51.7	47.6
	10%	WB	42.0	42.2	44.9	51.6	58.0	62.5	65.4	64.7	60.2	55.0	47.6	43.6
		MCDB	43.6	44.8	48.3	58.6	65.5	70.0	74.3	73.5	66.1	58.1	49.2	45.0
Mean Daily Temperature Range	5% DB	MDBR	8.1	9.6	12.8	17.5	18.7	17.8	17.6	17.9	15.7	12.5	8.8	7.6
		MCDBR	9.0	11.2	18.9	24.9	25.7	25.6	24.7	25.0	22.4	15.9	10.1	8.4
		MCWBR	8.1	8.5	12.4	14.2	14.0	13.0	10.8	10.7	11.5	10.5	8.0	7.8
	5% WB	MCDBR	8.9	10.8	16.9	22.2	22.8	22.4	22.4	21.4	19.6	14.6	9.2	8.4
		MCWBR	8.2	8.8	11.7	13.3	13.4	12.3	10.6	10.4	10.8	10.0	7.8	7.9
Clear-Sky Solar Irradiance	taub	0.315	0.338	0.370	0.397	0.392	0.390	0.409	0.406	0.374	0.359	0.336	0.308	
	taud	2.315	2.313	2.256	2.252	2.309	2.336	2.302	2.323	2.395	2.396	2.364	2.307	
	Ebn at Noon	202	237	253	263	271	272	264	258	252	227	193	179	
	Edh at Noon	19	26	33	38	38	37	38	35	29	23	18	16	
All-Sky Solar Radiation	RadAvg	174	378	767	1300	1641	1740	1619	1339	972	515	217	122	
	RadStd	21	58	78	169	190	131	209	102	98	69	25	10	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.90	N/A	N/A	N/A	+0.70	+0.97	-304	-430	+137	N/A
(47) Regional (52 neighbors)		+0.72	N/A	N/A	N/A	N/A	N/A	N/A	-295	N/A	N/A

Nomenclature: See separate page